ARTIFICIAL INTELLIGENCE – COURSE ASSIGNMENT

BUDGETBUDDY

SMART MONTHLY EXPENSE & SAVINGS AGENT

DESIGN OF AN AI AGENT – AGENT FORMULATION, ENVIRONMENT TYPE, SENSORS & ACTUATORS, AND PEAS FRAMEWORK EVALUATION

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BudgetBuddy

Smart Monthly Expense & Savings Agent

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1. Problem Statement

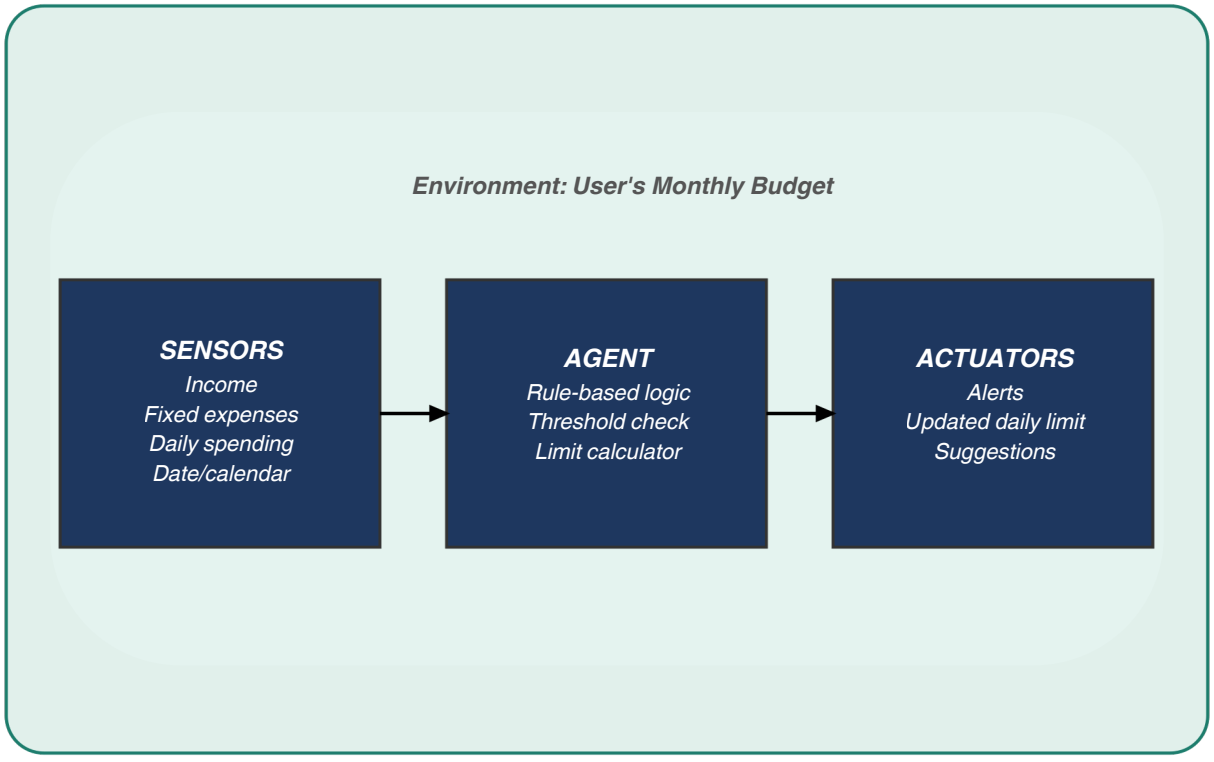
Students and young professionals often overspend their monthly budget and are unable to handle unexpected expenses toward the end of the month, mainly because manual budgeting does not adjust to real spending behaviour as the month progresses.

2. Proposed Agent: BudgetBuddy

BudgetBuddy takes monthly income,fixed expenses, and daily spending as input. It calculates a dynamic daily spending limit for the remaining days of the month, compares real spending against this limit, and generates alerts whenever a threshold is crossed. All decisions are made using simple condition-action rules.

3. Graphical Representation

The diagram shows how the agent senses financial data, processes it through rule-based logic, and acts on the environment through alerts and updated limits.



4. Agent Formulation

Agent Type

BudgetBuddy is a **Model-Based Reflex Agent**. It keeps track of an internal state (remaining budget and days left) and applies condition-action rules to decide when to alert the user.

Environment Type

Property	Type
Observability	Partially observable
Determinism	Deterministic
Episodic / Sequential	Sequential
Static / Dynamic	Dynamic
Discrete / Continuous	Discrete
Agents	Single-agent

Sensors

- Income entered by the user
- Fixed monthly expenses (rent, bills, subscriptions)
- Daily spending input
- System date (days remaining in month)

Actuators

- Spending alert / notification
- Updated dynamic daily spending limit
- Suggestions to reduce specific expense categories

5. PEAS Framework

Element	Description
Performance Measure	Days within budget, reduced overspending, alert accuracy, savings achieved.
Environment	User's monthly income, fixed and daily expenses, remaining days in the month.
Actuators	Alerts, updated daily limit, spending-reduction suggestions.
Sensors	Income entry, expense entries, date/calendar reader.